Lab 2 – TrashTag Product Specification

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1 Introduction

Every year, millions of pounds of trash are dumped all over American cities and towns. Over the past 36 years in Virginia alone, volunteers have removed approximately 7.1 million pounds of litter [1]. Large trash items, such as mattresses, old tires, and appliances are often dumped in recreational areas, which makes it difficult for visitors to attempt to remove them to an appropriate location. Illegal dumping has also led to about 100,000 marine animal deaths every year [2] and an estimated drop of property values by 7% to 10% in Hampton Roads [3]. Finally, when the general public witnesses illegal dumping or littering, they are more likely to litter in that area as well [4].

Since current solutions—such as proper volunteer efforts, communication between citizens and organizations, and disposal methods—are ineffective in the long term, citizens and organizations need a way to address the overarching issue. TrashTag is a web application that is designed to create a solution for this issue. The application itself gives the general public, volunteers, and organizations a way to properly and efficiently report and verify piles of trash in real time in order to help clean up local cities and recreational areas.

1.1 Purpose

TrashTag is a web application that aims to help citizens, non-profit organizations, and government agencies know where litter is located in their cities and help them to organize clean-up efforts. The main purpose of this System Requirements Specification (SRS) is to help guide the development team on what TrashTag is, how it works, what features to implement for it and how to implement those features. This document also defines what the functional and non-functional requirements are for the TrashTag application.

1.2 Scope

The goals and benefits of TrashTag are streamlining the current trash clean-up process and improving the visual look of local recreational and public spaces by locating and cleaning litter-affected areas. These benefits and goals will mostly be met via the key features of both the prototype and real-world product.

The system prototype will allow for citizens and organizations to view a live map, submit and track live litter reports, use a digital rewards system to track user's progress in real time, and verify the status of all reports. The prototype will also allow for citizens and organizations to take pictures of affected areas and upload those pictures to the report for verification and location tagging purposes. It will also allow organizations to properly allocate time and resources while organizing proper cleanup events in the local area for citizens to participate in.

The prototype application is not a replacement for emergency service reporting (e.g., toxic waste management). It also will not act as a law enforcement agency or penalize illegal littering or dumping. The prototype will also not guarantee a quick or accurate clean-up response, if one at all. Finally, the prototype will not provide users with tangible or monetary rewards for cleaning up trash or completing virtual gamification "milestones."

1.3 Definitions, Acronyms, and Abbreviations

API: Application Programming Interface. A software mechanism of accepting and providing data to and from external applications.

Database: An organized, electronic collection of data.

Environmental Organization: Private, and frequently non-profit, groups dedicated to environmental causes.

Geotagging: The process of adding GPS-provided location information onto images or videos taken with a camera.

Illegal Dumping: The dumping of trash in a non-authorized trash collection location, such as on the sidewalk or in a public park.

Microsoft Azure (Azure): Microsoft's Cloud computing platform, which offers over 200 products to its various business clients.

Non-Profit Organization: An organization that is private and not managed by the government.

Relational Database: A database that organizes data into distinct rows and columns for easy access.

SQL: Structured Query Language. A programming language used for storing and processing information in a relational database.

Trash (Litter): Unwanted or un(re)usable materials, such as food, paper, and plastic.

Web Application: An application software that runs on a web browser via the internet.

Web Browser (Browser): An application for accessing internet websites.

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1.5 Overview

This product specification includes the prototype's overall features and capabilities, major hardware and software components, user characteristics, constraints, and assumptions and dependencies.

2 Overall Description

The prototype for TrashTag will be developed as a web application that aims to improve the litter and waste cleanup process by allowing users to photograph and report piles of trash, give the location of the trash, and connect with cleanup groups that are well-equipped to handle the proper cleanup process. Most of the major functionalities will either be fully or partially implemented in the prototype, while the real-world product will have all functionalities fully implemented.

2.1 Product Perspective

The TrashTag prototype will utilize some third-party software to make the application function as intended. One such third-party software is OpenStreetMap's API, which contributes to the live litter reporting and mapping that uses the API. The prototype will have an architecture that is made out of three layers: the presentation layer, the application layer, and the data layer. The architecture design is shown in Figure 1.

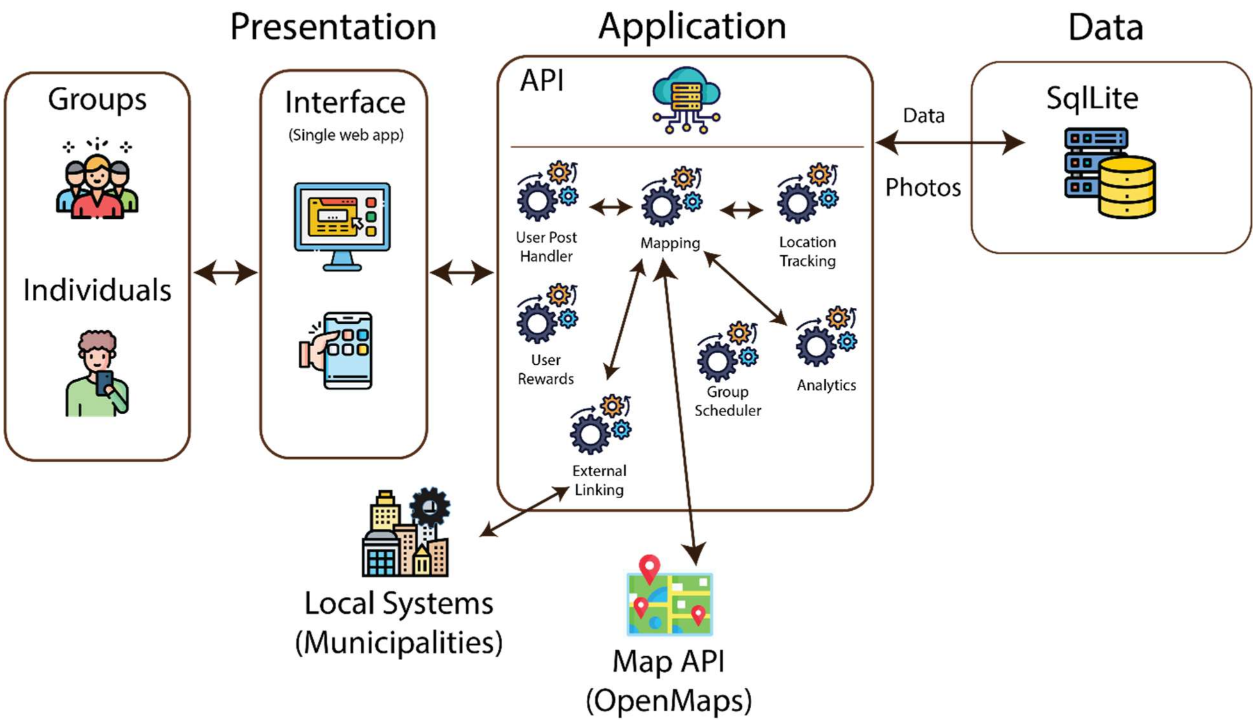

Figure 1

TrashTag's Major Functional Component Diagram (MFCD)

All three layers will work together to make the whole application functional. The presentation layer is created using frontend development languages, such as React and CSS. It will be where citizens and organizations interact with the application. The application layer is made up of the prototype’s backend, which is created using Python’s Django framework. Finally, the data layer is created using SQLite, which holds user, location, and report data.

2.2 Product Functions

In the TrashTag prototype, the main features of the software are its ability to report litter locations, upload photos, report the type of litter, mark the litter size in pounds, request cleanup verification status, view and filter a live report map, coordinate events for users, track cleanup status, verify cleanup status, view community impact statistics, view gamification rewards, view top contributors, utilize accessible resources, embed a map widget, provide analytics, moderate the platform, and authorize clients. All of these features and their implementation status (fully implemented, partially implemented, or eliminated) can be seen in Table 1.

Table 1

TrashTag Prototype Features and Implementation Status

Features	Prototype Implementation Status
Report Litter Location	Fully Implemented
Upload Photos	Fully Implemented
Report Litter Type	Fully Implemented
Mark Trash Size (lbs.)	Fully Implemented
Request Cleanup Verification Status	Fully Implemented
View Litter Map	Fully Implemented
View Litter Density Map	Eliminated
Filter by Status	Eliminated
Filter by Type	Partially Implemented
Filer by Location	Fully Implemented
Coordinate Cleanup Events	Partially Implemented
Claim Cleanup Areas	Partially Implemented
Delegate Cleanup Tasks	Partially Implemented
View Cleanup Events	Partially Implemented
RSVP to Events	Partially Implemented
Check-In to Events	Partially Implemented
Proximity Cleanup Alerts	Partially Implemented
Update Cleanup Status	Fully Implemented

Upload Cleanup Proof	Fully Implemented
Monitor Cleanup Status	Fully Implemented
Verify Accurate Cleanup Status	Fully Implemented
Verify Cleanup Completion	Fully Implemented
Flag Suspicious Content	Eliminated
View Impact Statistics	Partially Implemented
Gamification Rewards	Fully Implemented
Top Contributors Leaderboard	Fully Implemented
Community Guidelines	Partially Implemented
Embed Map Widget	Partially Implemented
Highlight Predicted Litter Trends	Eliminated
Download Reports	Eliminated
View Density-Heavy Litter Analytics	Eliminated
Track Event Impact	Eliminated
View Cleanup Statistics	Eliminated
Review Flagged Reports	Eliminated
Respond to Flagged Reports	Eliminated
Edit Flagged Report Data	Eliminated
Remove Suspicious Reports	Eliminated
Mark Reports as Spam	Eliminated
Issue Violation Warnings	Eliminated
Create New Account	Fully Implemented
Log Into Account	Fully Implemented

View and Edit User Profile	Fully Implemented

2.3 User Characteristics

The various user roles for the TrashTag prototype are citizens, organizations, local governments, and system moderators. The user expertise levels for citizens are users that have a basic knowledge of using web applications and accessing the internet on a mobile device or computer. Citizen usage scenarios are reporting litter-affected areas by uploading photos, collaborating with other users, and earning rankings as virtual rewards. Citizen constraints include users not being able to modify or aggregate any of the data on the platform.

The user expertise levels for organizations are users that have the same abilities as citizens, and also have the ability to organize cleanup events and group volunteers together. Organization usage scenarios include organizing and allocating proper resources to successfully clean affected areas. Organization constraints include not being able to view litter report and litter density trends and not being able to gather statistics from those trends.

The user expertise levels for local governments are those that have the ability to access statistics about litter reports and aggregate those statistics together. Local government usage scenarios include reducing cleanup costs and optimizing waste management resources. Constraints for local governments include not being able to view user information on the platform, as it remains anonymous to them.

Finally, user expertise levels for system moderators are for those that have the ability to ensure integrity and quality control over the application system. System moderator usage scenarios include being responsible for user account management, removing inappropriate and duplicate trash reports, and responding to community-flagged report submissions. The constraints on the system moderators are ensuring user safety, adhering to content policy guidelines, and enforcing regulations and legal compliance.

2.4 Constraints

The technical constraints for the TrashTag prototype are that it must run on a device that has access to the internet, access to location services, and runs either Windows, Mac, or Linux as its operating system. The legal constraints include that users must not use the prototype to trespass onto private property, organize crime, or dox other users on the platform. Users must also agree to the End-User License Agreement (EULA) upon registering a new account and joining the platform. The environmental constraints are that users must not use the application to endanger their local environments when they clean up affected areas.

2.5 Assumptions and Dependencies

The dependencies for TrashTag are OpenStreetMap's API, React, Python (Django), and SQLite, and Digital Ocean for the website hosting service.